

1 Problem Statement 1

P5-6c A reversible liquid-phase isomerization $A \rightleftharpoons B$ is carried out *isothermally* in a 1000-gal CSTR. The reaction is *second order in both the forward and reverse directions*. The liquid enters at the top of the reactor and exits at the bottom. Experimental data taken in a batch reactor shows the CSTR conversion to be 40%. The reaction is reversible with $K_C = 3.0$ at 300 K, and $\Delta H_{Rx}^\circ = -25,000$ cal/mol. Assuming that the batch data taken at 300 K are accurate and that $E = 15,000$ cal/mol, what CSTR temperature do you recommend to obtain maximum conversion? [Hint: Read Appendix C and assume $\Delta C_p = 0$ in the appendix Equation (C-9)]:

$$K_C(T) = K_C(T_0) \exp \left[\frac{\Delta H_{Rx}^\circ}{R} \left(\frac{1}{T_0} - \frac{1}{T} \right) \right]$$

Figure 1: Problem

Solution:

Code: Check the attached Matlab file.

Graph:

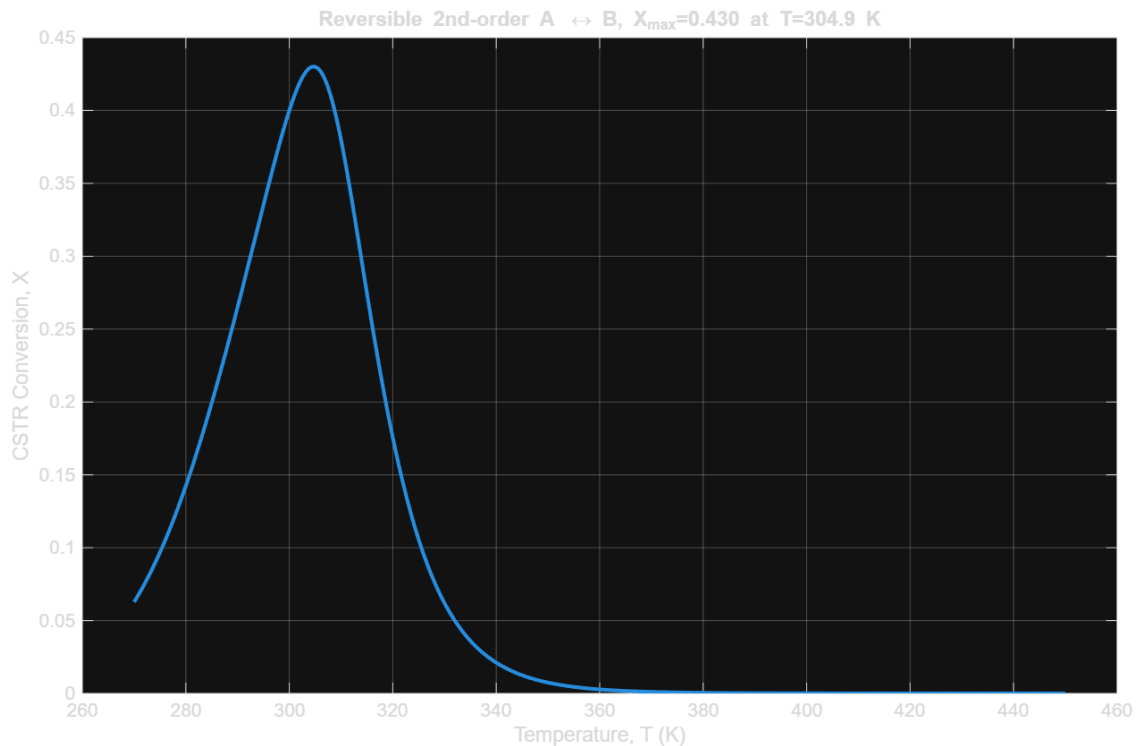


Figure 2: Problem

Does $X(T)$ go through a maximum? Why?

Yes. For an **exothermic** reversible reaction in a CSTR:

- Increasing T increases the forward rate constant k_1 (Arrhenius), which tends to increase X at low temperatures (kinetic-limited region).
- But increasing T **decreases** the equilibrium constant $K_c(T)$ (van't Hoff with $\Delta H_{\text{rxn}}^\circ < 0$), which lowers the thermodynamic limit

$$X_{\text{eq}} = \frac{K_c}{1 + K_c}.$$

At higher temperatures, the reactor becomes equilibrium-limited and X falls.

The competition between faster kinetics and less favorable equilibrium creates a single interior maximum in $X(T)$. The script above computes and shows that X peaks and its temperature.

2 Problem Statement 2

P5-9A The liquid-phase reaction



follows an elementary rate law and is carried out isothermally in a flow system. The concentrations of the A and B feed streams are 2 M before mixing. The volumetric flow rate of each stream is 5 dm³/min, and the entering temperature is 300 K. The streams are mixed immediately before entering. Two reactors are available. One is a gray, 200.0-dm³ CSTR that can be heated to 77°C or cooled to 0°C, and the other is a white, 800.0-dm³ PFR operated at 300 K that cannot be heated or cooled but can be painted red or black. Note that $k = 0.07 \text{ dm}^3/\text{mol} \cdot \text{min}$ at 300 K and $E = 20 \text{ kcal/mol}$.

- Which reactor and what conditions do you recommend? Explain the reason for your choice (e.g., color, cost, space available, weather conditions). Back up your reasoning with the appropriate calculations.
- How long would it take to achieve 90% conversion in a 200-dm³ batch reactor with $C_{A0} = C_{B0} = 1 \text{ M}$ after mixing at a temperature of 77°C?
- What would your answer to part (b) be if the reactor were cooled to 0°C? (*Ans.: 2.5 days*)
- What conversion would be obtained if the CSTR and PFR were operated at 300 K and connected in series? In parallel with 5 mol/min to each?
- Keeping Table 4-3 in mind, what batch reactor volume would be necessary to process the same amount of species A per day as the flow reactors, while achieving 90% conversion? Referring to Table 1-1, estimate the cost of the batch reactor.
- Write a couple of sentences describing what you learned from the problem and what you believe to be the point of the problem.
- Apply one or more of the six ideas in Preface Table P-4, page xxviii, to this problem.

Figure 3: Problem

(a) Reactor Selection

Arrhenius law:

$$k(T) = k(300) \exp \left[-\frac{E}{R} \left(\frac{1}{T} - \frac{1}{300} \right) \right], \quad R = 1.987 \text{ cal/mol-K.} \quad (1)$$

Results:

$$k(273 \text{ K}) = 2.59 \times 10^{-3}$$

$$k(300 \text{ K}) = 0.07$$

$$k(350 \text{ K}) = 8.45$$

CSTR design equation:

$$V = \frac{F_{A0}X}{kC_{A0}C_{B0}(1-X)^2} = \frac{10X}{k(1-X)^2}. \quad (2)$$

At $V = 200 \text{ dm}^3$:

$$X(0^\circ\text{C}) = 0.047,$$

$$X(300\text{K}) = 0.44,$$

$$X(77^\circ\text{C}) = 0.926.$$

PFR design:

$$X = \frac{\frac{kV}{v_0}}{1 + \frac{kV}{v_0}}. \quad (3)$$

At $V = 800 \text{ dm}^3$, $k(300\text{K}) = 0.07$: $X = 0.849$.

Conclusion: The 200 dm^3 CSTR at 77°C is optimal (92.6%).

(b) Batch Reactor at 77°C

Batch equation:

$$\frac{dX}{dt} = kC_{A0}(1-X)^2, \quad t = \frac{1}{kC_{A0}} \left[\frac{1}{1-X} - 1 \right]. \quad (4)$$

At $X = 0.9$, $k = 8.45$:

$$t = \frac{9}{8.45} = 1.07 \text{ min.} \quad (5)$$

(c) Batch time at 0°C to reach 90% conversion

Kinetics in a constant-volume batch, equimolar feed. For $A+B \rightarrow C$ with elementary rate law,

$$r_A = -k C_A C_B, \quad C_A = C_B = C_{A0}(1 - X) \quad (\text{equimolar, liquid, const. } v),$$

the mole balance gives

$$\frac{dC_A}{dt} = r_A \Rightarrow \frac{dX}{dt} = k C_{A0}(1 - X)^2.$$

Separate variables and integrate from $X = 0$ to X :

$$\int_0^X \frac{dX}{(1 - X)^2} = k C_{A0} \int_0^t dt \Rightarrow \left[\frac{1}{1 - X} \right]_0^X = k C_{A0} t,$$

$$\boxed{t(X) = \frac{1}{k C_{A0}} \left(\frac{1}{1 - X} - 1 \right)} = \frac{X}{k C_{A0}(1 - X)}.$$

Evaluate at $X = 0.90$, $T = 273 \text{ K}$. Use Arrhenius to get $k(273 \text{ K})$:

$$k(273) = k(300) \exp \left[-\frac{E}{R} \left(\frac{1}{273} - \frac{1}{300} \right) \right] = 0.07 \exp \left(-\frac{20000}{1.987} (0.003663 - 0.003333) \right) = 2.587 \times 10^{-3} \frac{\text{dm}^3}{\text{mol} \cdot \text{min}}$$

Then

$$t_{0^\circ\text{C}}(0.90) = \frac{0.90}{(2.587 \times 10^{-3})(1)(0.10)} = \frac{9}{2.587 \times 10^{-3}} = 3.48 \times 10^3 \text{ min}.$$

Convert to hours/days:

$$3.48 \times 10^3 \text{ min} \times \frac{1 \text{ h}}{60 \text{ min}} = 58.0 \text{ h} \approx \boxed{2.4\text{--}2.5 \text{ days}}.$$

(d) Conversion with CSTR & PFR at 300 K: series vs. parallel

Common data at 300 K

$$k = 0.07 \frac{\text{dm}^3}{\text{mol} \cdot \text{min}}, \quad F_{A0} = 10 \frac{\text{mol}}{\text{min}}, \quad v_0 = 10 \frac{\text{dm}^3}{\text{min}}.$$

Series: CSTR ($V = 200$) $\rightarrow$ PFR ($V = 800$) at 300 K. *Stage 1 (CSTR).* With $C_{A0} = C_{B0} = 1$,

$$V = \frac{F_{A0} X}{k(1 - X)^2} \Rightarrow 200 = \frac{10 X_1}{0.07(1 - X_1)^2} \Rightarrow X_1 = 0.4396.$$

Stage 2 (PFR). For second-order, constant v :

$$\int_{X_1}^{X_2} \frac{dX}{(1-X)^2} = \frac{kV_{\text{PFR}}}{v_0} \Rightarrow \frac{1}{1-X_2} - \frac{1}{1-X_1} = \frac{0.07 \times 800}{10} = 5.6.$$

Compute:

$$\frac{1}{1-X_1} = \frac{1}{1-0.4396} = 1.7847 \Rightarrow \frac{1}{1-X_2} = 1.7847 + 5.6 = 7.3847,$$

$$\boxed{X_{\text{series}} = X_2 = 1 - \frac{1}{7.3847} = 0.8646 \quad (86.5\%)}$$

Parallel: split feed equally (5 mol/min & 5 dm³/min per branch). CSTR branch ($V = 200$).

$$200 = \frac{F_{A0}^{(b)} X}{k(1-X)^2} = \frac{5X}{0.07(1-X)^2} \Rightarrow X_{\text{CSTR}} = 0.5548.$$

PFR branch ($V = 800$).

$$\frac{kV}{v_b} = \frac{0.07 \times 800}{5} = 11.2 \Rightarrow X_{\text{PFR}} = \frac{11.2}{1 + 11.2} = 0.9180.$$

Combine outlets (equal volumetric flows; same C_{A0}). Overall conversion is flow-weighted:

$$X_{\text{overall}} = \frac{v_b}{v_0} X_{\text{CSTR}} + \frac{v_b}{v_0} X_{\text{PFR}} = \frac{1}{2}(0.5548 + 0.9180) = \boxed{0.736 \quad (73.6\%)}$$

Remark. A single 1000-dm³ PFR at 300 K gives $X = \frac{kV/v_0}{1 + kV/v_0} = \frac{0.07 \times 1000/10}{1 + 0.07 \times 1000/10} = \frac{7}{8} = 87.5\%$, slightly higher than the series train because second-order kinetics favors plug flow.

(e) Batch volume to process the same moles of A per day at $X = 90\%$

Daily amount to process (from the flow system).

$$\dot{N}_{A,\text{day}} = F_{A0} \times 1440 = 10 \times 1440 = \boxed{14,400 \text{ mol/day}}$$

Reaction time to reach X in batch. From part (b)/(c):

$$t_b(X) = \frac{1}{kC_{A0}} \left(\frac{1}{1-X} - 1 \right) = \frac{X}{kC_{A0}(1-X)}.$$

At 77°C ($T = 350$ K), Arrhenius gives $k = 8.447$ dm³/mol·min; therefore, at $X = 0.90$,

$$t_b = \frac{0.90}{8.447 \times 1 \times 0.10} = \frac{9}{8.447} = 1.07 \text{ min.}$$

Idealized cycle (reaction time only). Number of batches/day:

$$N_{\text{batches}} = \frac{1440}{t_b} = \frac{1440}{1.07} = 1346.$$

Each batch initially contains $C_{A0}V_b = 1 \times V_b$ mol of A. Material balance per day:

$$N_{\text{batches}} V_b = \dot{N}_{A,\text{day}} \Rightarrow \boxed{V_b = \frac{14,400}{1346} = 10.7 \text{ dm}^3 \text{ (10.7 L).}}$$

More realistic cycle with overhead. If nonreaction operations take t_{nr} minutes per batch,

$$t_{\text{cycle}} = t_b + t_{\text{nr}}, \quad V_b = \frac{\dot{N}_{A,\text{day}}}{C_{A0}} \cdot \frac{t_{\text{cycle}}}{1440}.$$

For $t_{\text{nr}} = 4$ min $\Rightarrow t_{\text{cycle}} = 5$ min:

$$N_{\text{batches}} = \frac{1440}{5} = 288, \quad \boxed{V_b = \frac{14,400}{288} = 50 \text{ L.}}$$

Cost note (order of magnitude). Small jacketed glass reactors (10–50 L) typically range from a few thousand to $\sim$ \$10–\$15k depending on jacket, drive, and thermal unit. Stainless sanitary vessels can be cheaper used; add utilities (heater/chiller) as needed.

(f) The problem implies

1. **Temperature control dominates performance.** For irreversible kinetics, raising T can more than offset larger volume; here a 200-dm³ CSTR at 77°C outperformed an 800-dm³ PFR at 300 K.
2. **Reactor ordering matters.** With second-order kinetics ($r \propto C^2$), PFRs are inherently more efficient than CSTRs at a given volume; splitting flow in parallel can reduce overall conversion.
3. **Batch vs. flow trade-offs.** When kinetics are fast (here $t_{90\%} \approx 1$ min at 77°C), batch processing can match continuous throughput with modest working volumes, but cycle overheads quickly increase the required size.

(g) “What if ideas...”

- **Non-stoichiometric feeds:** Let $C_{B0} = \alpha C_{A0}$ with $\alpha > 1$. Re-derive CSTR/PFR formulae with $C_A = C_{A0}(1 - X)$, $C_B = C_{B0} - C_{A0}X$. Show that modest B excess increases rate and reduces required volume for a target X .
- **Equilibrium effects:** If the reaction is reversible and exothermic with $K_c(T)$ from van 't Hoff, plotting $X(T)$ for a CSTR exhibits a *maximum* as the Arrhenius speed-up (kinetics) competes with decreasing K_c (thermodynamics).
- **Heat removal limits:** Impose a maximum coolant duty; solve for attainable T and hence X . This may flip the choice back toward the isothermal PFR if the CSTR cannot be maintained at 77°C at scale.
- **Scale-up:** Repeat parts (a–e) for 10× throughput; include pressure drop in the PFR and agitator power in the CSTR.